

The page title the retriever never saw

Hebrew RAG over Kol-Zchut · retrieval stage only · 24,487 paragraphs, 7,007 pages

42%

of questions send the LLM three pages that do not contain the right one (baseline hit@3 = 0.58; hit@1 = 0.37)

1

The retriever embeds the paragraph body only.

The page title is stored next to it but never reaches the model.

2

63 of 187 misses are rank 2-3, mostly sibling pages.

ביומטרי vs לנכי עבודה · דרכון זמני vs הנחה בארנונה לנכים
a page vs its wartime twin

3

64% of paragraphs already start with their section heading.

So the section is in the vector; the page is the missing piece.

Two config keys put the page title into the vector

1 `embed_context_fields: ["title"]`

Text embedded at indexing time becomes "`<page title>\n<paragraph>`"

2 `embed_title_weight: 0.3`

Title embedded on its own and fused into the stored vector:

`normalise(0.3·unit(title) + 0.7·unit(text))`

Cosine is linear in the query, so one stored vector = $0.3 \cdot \cos(q, \text{title}) + 0.7 \cdot \cos(q, \text{text})$. Same model, same index layout, same /search.

1

Measure first

Harness on cached vectors, identical to Webiks' evaluator to 4 decimals; 3 s per variant.

2

Read the errors

Sibling confusions → the title is where the qualifier lives.

3

Predict, then test

Written before the run; half right (deep misses moved too). Kept as written.

4

Choose weight honestly

Swept on half the questions, reported on the other half, both ways; plateau 0.2–0.4.

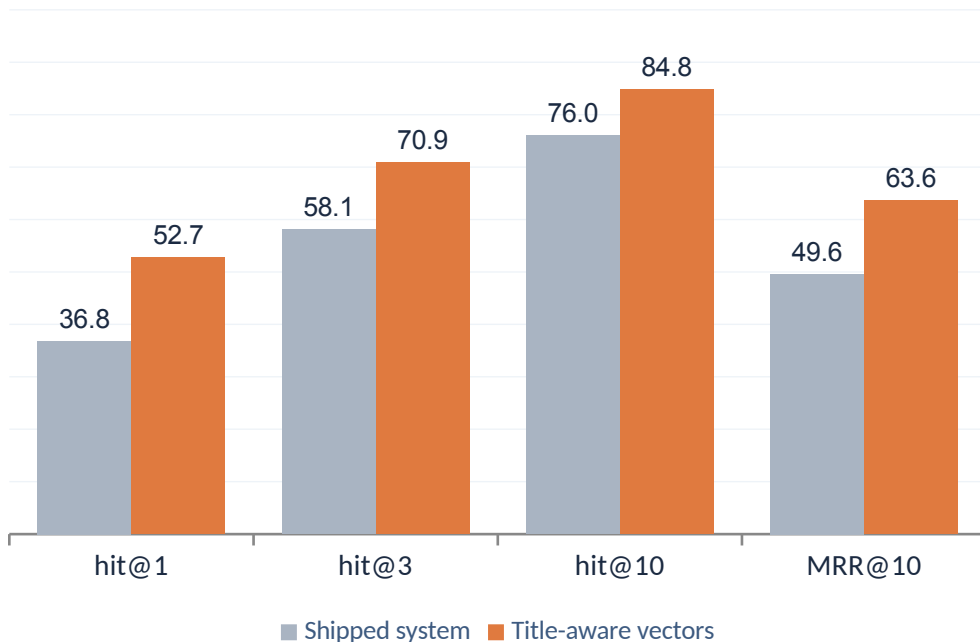
5

Reject on evidence

Chunking, LLM-written context, boilerplate filter, wider pool: all measured, none shipped.

Rank 1: 37% → 53% · LLM's top 3: 58% → 71%

296 held-out questions, full corpus (%)



Per question

138 improved · 35 worse · 123 unchanged. Most regressions move one place between sibling pages.

Tested, not shipped

- 256-token chunks: worse alone, trades rank-1 for top-3 with the title
- LLM-written context per paragraph (\$3.34): every metric down
- Boilerplate filter, wider candidate pool: no effect

Next

Small-to-big retrieval (match chunks, return the paragraph) · a human-written clean eval set · a reranker as a measured second stage